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CUET UG

Physics (322)

Magnetic Effects of Current & Magnetism

Official Syllabus Unit 3 · Class 12 NCERT · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA Physics-322, Unit 3)	Key Exam Focus & Skills
Biot-Savart Law & Circular Loops	Concept of magnetic field, Oersted's experiment. Biot-Savart law and its application to the current carrying circular loop.	Vector cross product $d(B) \propto Id(l) \times \hat{r}$, axial field $B_x \propto \frac{R^2}{(R^2+x^2)^{\frac{3}{2}}}$, center field $\frac{\mu_0 NI}{2R}$, right-hand thumb rule.
Ampere's Law & Solenoids	Ampere's law and its applications to infinitely long straight wire. The solenoid.	Line integral $\oint (B) \cdot d(l) = \mu_0 I_{\text{enc}}$, long wire $\frac{\mu_0 I}{2\pi r}$, solenoid core $\mu_0 nI$ vs end $\frac{1}{2}\mu_0 nI$.
Lorentz Force & Particle Dynamics	Force on a moving charge in uniform magnetic and electric fields.	$(F) = q((E) + (v) \times (B))$, zero magnetic work $(\Delta K = 0)$, gyroradius $r = \frac{mv}{qB}$, helical pitch, velocity selector $v = \frac{E}{B}$.
Force on Conductors & Dipoles	Force on a current-carrying conductor in a uniform magnetic field, force between two parallel current-carrying conductors - definition of ampere, current loop as a magnetic dipole and its magnetic dipole moment, torque experienced by a current loop in uniform magnetic field.	$(F) = I(L) \times (B)$, parallel wires force per unit length $\frac{\mu_0 I_1 I_2}{2\pi d}$ (like attract, unlike repel), dipole moment $(M) = NI(A)$, torque $(\tau) = (M) \times (B)$.
Moving Coil Galvanometer	Moving coil galvanometer - its current sensitivity and conversion to ammeter and voltmeter.	Radial magnetic field, linear deflection $\theta \propto I$, current sensitivity $NA\frac{B}{C}$, ammeter shunt $S = \frac{I_g G}{I - I_g}$, voltmeter multiplier $R_s = \left(\frac{V}{I_g}\right) - G$.
Bar Magnet & Magnetic Materials	Bar magnet, bar magnet as an equivalent solenoid, magnetic field lines, magnetic field intensity due to a magnetic dipole along its axis and perpendicular to its axis, torque on a magnetic dipole in a uniform magnetic field; Magnetic properties of materials - Para-, dia- and ferro-magnetic substances with examples, magnetization of materials, effect of temperature on magnetic properties.	Axial $(2k_m \frac{M}{r^3})$ vs equatorial $(k_m \frac{M}{r^3})$, Gauss's law for magnetism $(\oint (B) \cdot d(A) = 0)$, susceptibility χ_m, Curie's law $\chi \propto \frac{1}{T}$, hysteresis loop retentivity & coercivity.

EXAM PATTERN

CUET UG Exam Architecture: Unit 3 (Magnetic Effects of Current and Magnetism) constitutes approximately 10% to 12% of the CUET UG Physics test (5 to 6 MCQs). High-frequency question archetypes include cyclotron trajectory variables, ammeter/voltmeter range extensions, force between parallel currents, and temperature behavior of magnetic susceptibilities.

Concept of Magnetic Field & Biot-Savart Law

Oersted's Discovery & Magnetic Field Concept

In 1820, Hans Christian Oersted demonstrated that an electric current deflects a nearby magnetic needle, establishing that moving electric charges generate a magnetic field.

- A static electric charge produces only an electrostatic field (E).
- A steadily moving electric charge produces both an electric field (E) and a magnetic field (B).
- An accelerated charge emits electromagnetic radiation.

Biot-Savart Law (Differential Field Element)

Jean-Baptiste Biot and Félix Savart quantitatively formulated the magnetic field $d(B)$ produced at a point P with position vector (r) by an infinitesimal current element $Id(l)$:

BIOT-SAVART LAW (VECTOR FORMULATION)

$$d(B) = \left(\frac{\mu_0}{4\pi}\right)\frac{Id(l) \times \hat{r}}{r^2} = \left(\frac{\mu_0}{4\pi}\right)\frac{Id(l) \times (r)}{r^3}$$

In scalar magnitude form:

$$dB = \left(\frac{\mu_0}{4\pi}\right)\frac{Idl\sin\theta}{r^2} \quad [\text{SI unit: Tesla (T) or Wb/m}^2]$$

where θ is the angle between the current element $d(l)$ and the displacement vector (r).

Constants & Permittivity-Permeability Relation:

- Permeability of free space: $\mu_0 = 4\pi \times 10^{-7}\ \text{T} \cdot \text{m} \cdot \text{A}^{-1} = 4\pi \times 10^{-7}\ \text{N} \cdot \text{A}^{-2}$.
- Constant factor: $\frac{\mu_0}{4\pi} = 10^{-7}\ \text{T} \cdot \text{m/A}$.
- Fundamental link connecting electromagnetism and optics:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 3.0 \times 10^8\ \text{m/s}$$

Magnetic Field of a Circular Current-Carrying Loop

Consider a circular loop of radius R carrying a steady current I.

1. **Magnetic Field on the Axis (distance x from center):** By resolving $d(B)$ into axial and transverse components, the transverse components cancel due to circular symmetry, leaving the net axial field:

$$B_{\text{axial}} = \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{\frac{3}{2}}}$$

2. **At the Center of the Circular Loop ($x = 0$):**

$$B_{\text{center}} = \frac{\mu_0 N I}{2R}$$

3. **Far Field along the Axis ($x \gg R$):**

$$B_{\text{axial}} \approx \frac{\mu_0 N I R^2}{2x^3} = \left(\frac{\mu_0}{4\pi}\right)\frac{2(NI\pi R^2)}{x^3} = \left(\frac{\mu_0}{4\pi}\right)\frac{2M}{x^3}$$

where $M = NIA = NI(\pi R^2)$ is the magnetic dipole moment of the circular loop.

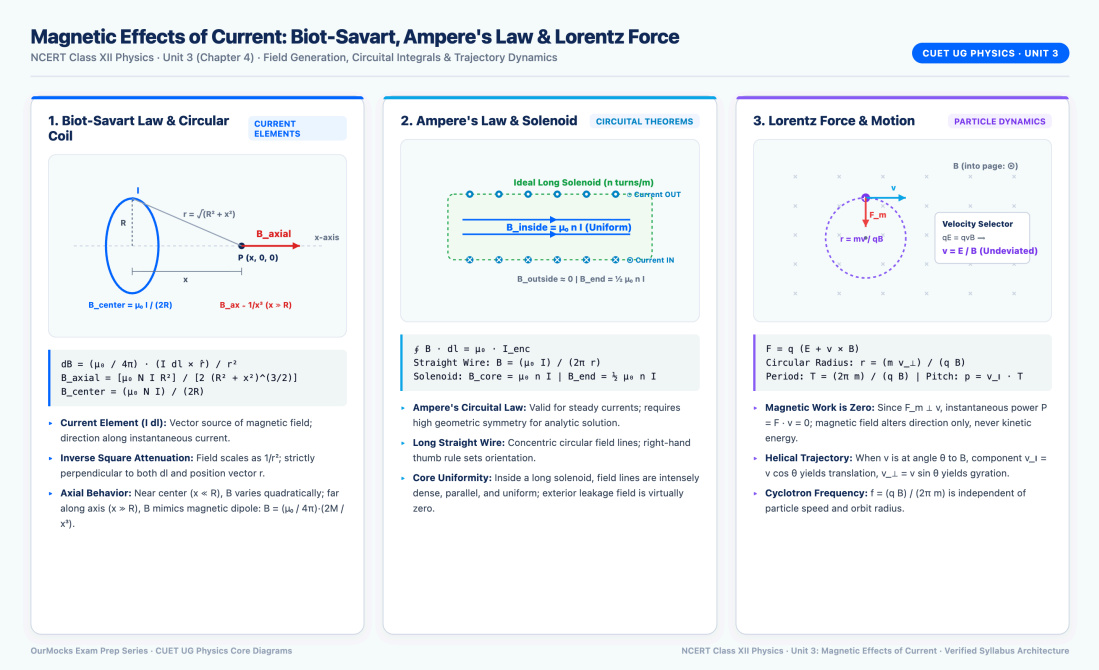


Figure 1: Electromagnetic Dynamics: Biot-Savart axial field, Ampere's circuital solenoid core, and Lorentz gyration.

WORKED EXAMPLE 1: CIRCULAR LOOP AXIAL VS CENTER RATIO

Problem: At what distance x along the axis of a circular current-carrying loop of radius R does the magnetic field fall to $\frac{1}{8}$th of its value at the center?

Solution:

1. Axial field formula: $B_{\text{axial}} = \frac{\mu_0 I R^2}{2(R^2+x^2)^{\frac{3}{2}}}$.
2. Center field formula: $B_{\text{center}} = \frac{\mu_0 I}{2R}$.
3. Given $\frac{B_{\text{axial}}}{B_{\text{center}}} = \frac{1}{8}$:

$$\frac{\frac{R^2}{(R^2+x^2)^{\frac{3}{2}}}}{\frac{1}{R}} = \frac{1}{8} \Longrightarrow \frac{R^3}{(R^2+x^2)^{\frac{3}{2}}} = \frac{1}{8}$$

4. Take reciprocal and raise to power $\frac{2}{3}$:

$$\left(\frac{R^2+x^2}{R^2}\right)^{\frac{3}{2}} = 8 \Longrightarrow \frac{R^2+x^2}{R^2} = 8^{\frac{2}{3}} = 4$$

5. Solve for x:

$$1 + \frac{x^2}{R^2} = 4 \Longrightarrow \frac{x^2}{R^2} = 3 \Longrightarrow x = \sqrt{3}R$$

Ampere's Circuital Law & Solenoid Fields

Ampere's Circuital Law (Integral Form)

Ampere's Circuital Law states that the line integral of the magnetic field (B) around any closed imaginary loop (Amperian loop) equals μ_0 times the total steady electric current enclosed by the loop:

AMPERE'S CIRCUITAL LAW

$$\oint (B) \cdot d(l) = \mu_0 I_{\text{enclosed}} = \mu_0 \sum I_{\text{in}}$$

Sign convention: Curl fingers of right hand along the direction of Amperian loop integration; current flowing in direction of thumb is taken as positive.

Applications of Ampere's Circuital Law

1. **Infinitely Long Straight Current-Carrying Conductor:** By choosing a circular Amperian loop of radius r centered on the wire:
$$\oint (B) \cdot d(l) = B(2\pi r) = \mu_0 I \Longrightarrow B = \frac{\mu_0 I}{2\pi r}$$
2. **Inside a Long Solid Cylindrical Wire of Radius a carrying uniform current I:**
   - For $r \geq a$ (outside): $B = \frac{\mu_0 I}{2\pi r} \propto \frac{1}{r}$.
   - For $r \leq a$ (inside): Enclosed current $I_{\text{enc}} = I\frac{\pi r^2}{\pi a^2} = I\left(\frac{r^2}{a^2}\right)$.
   $$B(2\pi r) = \mu_0 I\left(\frac{r^2}{a^2}\right) \Longrightarrow B = \left(\frac{\mu_0 I}{2\pi a^2}\right) r \propto r$$
   - **Field Profile:** B increases linearly from zero at axis to maximum at surface ($r = a$), then decays inversely as $\frac{1}{r}$.
3. **Ideal Long Solenoid (n turns per unit length):**
   - Inside the core: Field lines are uniform, parallel, and dense:
   $$B_{\text{inside}} = \mu_0 n I = \mu_0 \left(\frac{N}{L}\right) I$$
   - Near the ends of a semi-infinite solenoid:
   $$B_{\text{end}} = \frac{1}{2}\mu_0 n I$$
   - Outside an ideal long solenoid: $B_{\text{outside}} \approx 0$.
4. **Toroid (Endless Solenoid):** For a toroidal coil of mean radius r having total N turns:
$$B = \frac{\mu_0 N I}{2\pi r} = \mu_0 n I$$

WORKED EXAMPLE 2: SOLENOID TURNS & CURRENT

Problem: A solenoid of length 0.5 m has 500 turns of wire and carries a current of 2.5 A. Find the magnetic field (a) at the center of the solenoid and (b) near one of its ends.

Solution:

1. Number of turns per unit length: $n = \frac{N}{L} = \frac{500}{0.5} = 1000$ turns/m.
2. Center field:
$$B_{\text{center}} = \mu_0 n I = (4\pi \times 10^{-7}) \times 1000 \times 2.5 = 10\pi \times 10^{-4} \approx 3.14 \times 10^{-3} \text{ T}$$
3. End field:
$$B_{\text{end}} = \frac{1}{2} B_{\text{center}} = \frac{1}{2}(3.14 \times 10^{-3}) \approx 1.57 \times 10^{-3} \text{ T}$$

Lorentz Force & Trajectories of Charged Particles

Lorentz Force Equation

When a particle carrying electric charge q moves with velocity (v) in a region with both an electric field (E) and a magnetic field (B), the net electromagnetic force (Lorentz force) is:

$$(F) = (F)_e + (F)_m = q(E) + q((v) \times (B))$$

CHARACTERISTICS OF MAGNETIC LORENTZ FORCE

- **Magnitude:** $F_m = qvB\sin\theta$, where θ is the angle between (v) and (B).
- **Direction:** Given by right-hand screw rule for positive q (reversed for negative q).
- **Zero Force Conditions:** $F_m = 0$ if $q = 0$ (neutral particle), $v = 0$ (stationary charge), or $(v) \parallel (B)$ ($\theta = 0°$ or $180°$).
- **No Work Theorem:** Because $(F)_m \perp (v)$ at every instant:
$$P = (F)_m \cdot (v) = q((v) \times (B)) \cdot (v) = 0 \implies W = \int P dt = 0$$
A magnetic field can change the direction of motion of a charged particle, but it can **never change its speed or kinetic energy**.

Motion of a Charged Particle in a Uniform Magnetic Field

1. **Case 1: Velocity Perpendicular to Field ($(v) \perp (B)$, $\theta = 90°$):** The constant normal force $F_m = qvB$ provides centripetal acceleration, executing uniform circular motion:
$$\frac{mv^2}{r} = qvB \implies r = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mK}}{qB}$$
   - **Time Period & Cyclotron Frequency:**
$$T = \frac{2\pi r}{v} = \frac{2\pi m}{qB} \quad \text{and} \quad f_c = \frac{1}{T} = \frac{qB}{2\pi m}$$
   Crucially, T and f_c are **independent of the speed v and radius r** of the particle.
2. **Case 2: Velocity at an Arbitrary Angle θ to (B):** Resolve velocity into components:

- Parallel component: $v_{\parallel} = v\cos\theta$ (unaffected by (B), provides uniform axial translation).
- Perpendicular component: $v_{\perp} = v\sin\theta$ (generates circular gyration).

The resultant trajectory is a **helix**:

- Radius: $r = \frac{mv_{\perp}}{qB} = \frac{mv\sin\theta}{qB}$.
- Pitch of the helix (distance advanced along field per turn):

$$p = v_{\parallel}T = (v\cos\theta)\left(\frac{2\pi m}{qB}\right) = \frac{2\pi mv\cos\theta}{qB}$$

Velocity Selector (Crossed Fields)

When mutually perpendicular uniform electric and magnetic fields are arranged $((E)\perp(B))$, a charged particle projected perpendicular to both experiences opposing electrostatic and magnetic forces:

$$F_e = qE \quad \text{and} \quad F_m = qvB$$

When balanced ($qE = qvB$), the particle traverses undeviated with velocity:

$$v = \frac{E}{B}$$

Particles with $v = \frac{E}{B}$ emerge through the exit slit regardless of their charge or mass.

WORKED EXAMPLE 3: PROTON AND ALPHA PARTICLE TRAJECTORY

Problem: A proton and an α-particle enter a uniform transverse magnetic field with identical kinetic energies. Determine the ratio of their orbit radii $\frac{r_p}{r_\alpha}$.

Solution:

1. Orbital radius in terms of kinetic energy: $r = \frac{\sqrt{2mK}}{qB}$.
2. For identical K and B: $r \propto \frac{\sqrt{m}}{q}$.
3. Physical constants: $m_\alpha = 4m_p$ and $q_\alpha = 2q_p$.
4. Calculate ratio:

$$\frac{r_p}{r_\alpha} = \frac{\frac{\sqrt{m_p}}{q_p}}{\frac{\sqrt{m_\alpha}}{q_\alpha}} = \sqrt{\frac{m_p}{4m_p}} \times \left(2\frac{q_p}{q_p}\right) = \left(\frac{1}{2}\right) \times 2 = 1$$

5. The orbit radii are identical ($\frac{r_p}{r_\alpha} = 1$).

Forces on Conductors & Magnetic Dipole Dynamics

Magnetic Force on a Current-Carrying Conductor

A conductor of length (L) carrying current I in a uniform magnetic field (B) experiences a net force equal to the vector sum of Lorentz forces on all conduction electrons:

$$(F) = I((L)\times(B)) \quad \Longrightarrow \quad F = ILB\sin\theta$$

The direction of (F) is given by Fleming's Left-Hand Rule or the right-hand vector cross product.

Force Between Two Parallel Current-Carrying Conductors

Consider two long, parallel straight wires separated by distance d in vacuum carrying steady currents I_1 and I_2:

- Wire 1 generates magnetic field at wire 2: $B_1 = \frac{\mu_0 I_1}{2\pi d}$.
- Force experienced by length L of wire 2: $F_2 = I_2 L B_1 = \frac{\mu_0 I_1 I_2 L}{2\pi d}$.
- **Force per unit length ($f = \frac{F}{L}$):**

$$f = \frac{\mu_0 I_1 I_2}{2\pi d} \quad \text{[SI unit: N/m]}$$

CURRENT DIRECTION INVARIANT & DEFINITION OF AMPERE

- **Currents in same direction (parallel):** Conductors **attract** each other.
- **Currents in opposite directions (antiparallel):** Conductors **repel** each other.
- **Standard Definition of the Ampere:** One Ampere is that steady current which, when maintained in two straight parallel conductors of infinite length and negligible cross-section placed 1 meter apart in vacuum, produces between them a force equal to 2×10^{-7} N per meter of length.

Current Loop as a Magnetic Dipole

A planar loop of area A carrying current I acts as a magnetic dipole with magnetic dipole moment (M):

$$(M) = NI(A) \quad \text{[SI unit: A} \cdot \text{m}^2 = \text{J/T]}$$

The direction of (A) is defined by the right-hand grip rule (fingers curl along current, thumb points along (M)).

- **Magnetic Moment of an Orbiting Electron (Bohr Magneton):** For an electron revolving in an atom with radius r and speed v, orbital magnetic moment is:

$$\mu_l = \frac{1}{2} evr = \frac{e}{2m_e} L_z$$

where $L_z = m_e vr$ is orbital angular momentum. By Bohr's quantization ($L = n\frac{h}{2\pi} = n\frac{h}{2\pi}$), the minimum non-zero magnetic moment (Bohr Magneton μ_B) is:

$$\mu_B = \frac{e\frac{h}{2\pi}}{2m_e} = \frac{eh}{4\pi m_e} \approx 9.274 \times 10^{-24} \text{ A} \cdot \text{m}^2$$

Torque on a Current Loop in a Uniform Field

When a planar coil of area (A) carrying current I is placed in a uniform magnetic field (B):

$$(\tau) = (M) \times (B) \quad \Longrightarrow \quad \tau = MB\sin\theta = NIAB\sin\theta$$

where θ is the angle between magnetic moment (M) (normal to loop) and field (B).

- Potential energy of dipole: $U = -(M) \cdot (B) = -MB\cos\theta$.
- Stable equilibrium at $\theta = 0°$ ($\tau = 0, U_{\min} = -MB$).
- Unstable equilibrium at $\theta = 180°$ ($\tau = 0, U_{\max} = +MB$).

WORKED EXAMPLE 4: FORCE BETWEEN PARALLEL WIRES

Problem: Two long parallel wires separated by 4.0 cm carry currents of 6.0 A and 2.0 A in opposite directions. Find the magnitude and nature of the force per unit length acting on each wire.

Solution:

1. Force per unit length formula: $f = \frac{\mu_0 I_1 I_2}{2\pi d}$.
2. Substitute $I_1 = 6.0$ A, $I_2 = 2.0$ A, and $d = 0.04$ m:

$$f = \frac{(4\pi \times 10^{-7}) \times 6.0 \times 2.0}{2\pi \times 0.04} = \frac{2 \times 10^{-7} \times 12.0}{0.04} = \frac{24 \times 10^{-7}}{0.04} = 6.0 \times 10^{-5} \text{ N/m}$$

3. Since the currents are antiparallel, the force is strictly **repulsive**.

Moving Coil Galvanometer & Circuit Measuring Instruments

The moving coil galvanometer (MCG) is an ultrasensitive electromechanical instrument used to detect and measure minute electric currents ($\approx 10^{-9}$ A).

Construction & Principle of Radial Magnetic Field

An MCG consists of a rectangular coil wound over a non-magnetic metallic frame suspended between the cylindrical concave magnetic pole pieces of a strong permanent magnet:

- **Soft Iron Cylindrical Core:** Placed symmetrically inside the coil to:
  1. Intensify the magnetic field magnitude by its high magnetic permeability ($\mu_r \gg 1$).
  2. Make the magnetic field strictly **radial** across the air gap.
- **Significance of Radial Field:** In a radial field, the plane of the coil is always parallel to magnetic field lines ($\theta = 90°$) regardless of coil orientation.

$$\tau_d = NIAB \sin 90° = NIAB = \text{constant with respect to deflection angle}$$

Deflection & Sensitivity Relations

The deflecting torque τ_d is opposed by the restoring elastic torque $\tau_r = C\theta$ provided by the suspension strip (phosphor bronze):

$$NIAB = C\theta \Longrightarrow \theta = \left(\frac{NAB}{C}\right) I$$

where C is the restoring torque per unit twist (torsional stiffness).

- **Linear Scale:** Because $\theta \propto I$, the galvanometer possesses an evenly graduated linear scale.
- **Current Sensitivity (I_s):** Deflection produced per unit electric current:

$$I_s = \frac{\theta}{I} = \frac{NAB}{C} \quad \text{[SI unit: div/A or rad/A]}$$

- **Voltage Sensitivity (V_s):** Deflection produced per unit applied voltage across coil of resistance G:

$$V_s = \frac{\theta}{V} = \frac{\theta}{IG} = \frac{NAB}{CG} = \frac{I_s}{G} \quad \text{[SI unit: div/V or rad/V]}$$

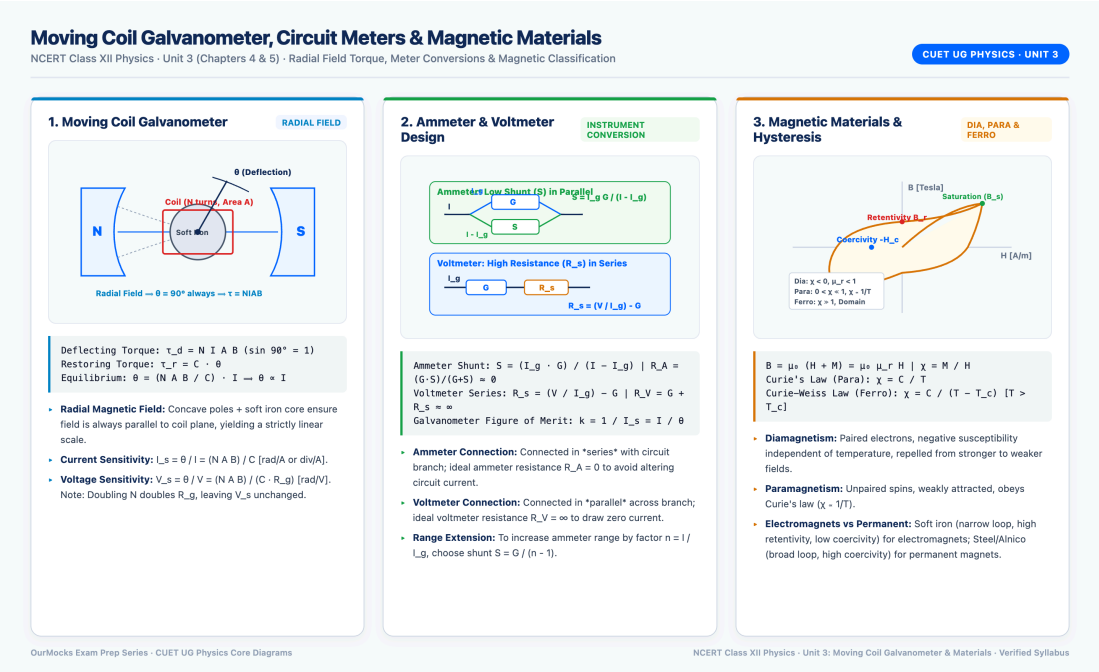


Figure 2: Moving Coil Galvanometer Architecture: Radial field torque, conversion to ammeter/voltmeter, and material hysteresis loop.

CRUCIAL MCQ TRAP: INCREASING CURRENT SENSITIVITY VS VOLTAGE SENSITIVITY

Increasing the number of turns N increases current sensitivity ($I_s \propto N$). However, adding turns lengthens the wire, proportionally increasing the internal resistance ($G \propto N$). Consequently, the ratio $V_s = \frac{I_s}{G}$ remains **unchanged**. Thus, increasing current sensitivity does **not** necessarily increase voltage sensitivity.

Conversion of Galvanometer into Ammeter and Voltmeter

Parameter	Ammeter Conversion	Voltmeter Conversion
Purpose	Measures steady branch current I	Measures potential difference V
Circuit Connection	Connected in **series** with circuit branch	Connected in **parallel** across branch
Required Modification	Low resistance shunt S connected in **parallel**	High resistance multiplier R_s connected in **series**
Component Formula	$I_g G = (I - I_g)S$ $S = \frac{I_g G}{I - I_g} \approx \frac{G}{n}$	$V = I_g(G + R_s)$ $R_s = \frac{V}{I_g} - G$
Equivalent Resistance	$R_A = \frac{GS}{G+S} \approx S$ (very small) **Ideal Ammeter:** $R_A = 0$	$R_V = G + R_s$ (very large) **Ideal Voltmeter:** $R_V = \infty$

WORKED EXAMPLE 5: SHUNT AND MULTIPLIER CALCULATIONS

Problem: A galvanometer of coil resistance $G = 20\Omega$ gives full-scale deflection for a current of $I_g = 5$ mA. Calculate (a) the shunt S needed to convert it into an ammeter of range $0 - 5$ A, and (b) the series resistance R_s to convert it into a voltmeter of range $0 - 10$ V.

Solution:

1. (a) Shunt resistance for ammeter ($I = 5$ A, $I_g = 0.005$ A):

$$S = \frac{I_g G}{I - I_g} = \frac{0.005 \times 20}{5 - 0.005} = \frac{0.1}{4.995} \approx 0.020\Omega$$

2. (b) Series resistance for voltmeter ($V = 10$ V):

$$R_s = \frac{V}{I_g} - G = \frac{10}{0.005} - 20 = 2000 - 20 = 1980\Omega$$

Magnetism of Matter & Earth's Magnetic Field

Bar Magnet as an Equivalent Solenoid

A bar magnet of length $2l$ and pole strength m has magnetic dipole moment $M = m(2l)$. It behaves identically to a finite current-carrying solenoid of equal dipole moment:

- **Axial Field ($\theta = 0°$):** For $r \gg l$:

$$B_{\text{axial}} = \left(\frac{\mu_0}{4\pi}\right)\frac{2M}{r^3}$$

- **Equatorial Field ($\theta = 90°$):** For $r \gg l$:

$$B_{\text{equatorial}} = \left(\frac{\mu_0}{4\pi}\right)\frac{M}{r^3}$$

- **Ratio:** $\frac{B_{\text{axial}}}{B_{\text{equatorial}}} = 2$ at identical distances.

Gauss's Law for Magnetism

GAUSS'S LAW FOR MAGNETISM

The net magnetic flux emerging through any arbitrary closed Gaussian surface is identically zero:

$$\Phi_B = \oint (B) \cdot d(A) = 0$$

Physical Meaning: Isolated magnetic monopoles do **not exist** in nature; magnetic field lines are continuous closed loops with neither beginning nor end.

Classification of Magnetic Materials

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Electronic Origin	Paired electrons; induced orbital dipole opposing field	Unpaired electron spins; partial alignment with field	Exchange coupling forming magnetized magnetic domains
Susceptibility (χ_m)	Negative, small (-10^{-5}), independent of T	Positive, small ($+10^{-3}$), obeys Curie's law	Positive, very large (10^3 to 10^5), domain alignment

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Relative Permeability (μ_r)	$\mu_r < 1$ (slightly less than 1)	$\mu_r > 1$ (slightly greater than 1)	$\mu_r \gg 1$ (hundreds to thousands)
External Field Effect	Expelled from strong to weak field regions	Weakly pulled from weak to strong field	Strongly attracted into strongest field regions
Representative Examples	Bismuth, Copper, Water, Lead, Silicon	Aluminum, Sodium, Platinum, Oxygen (O_2)	Iron, Cobalt, Nickel, Alnico, Gadolinium

Temperature Dependence & Hysteresis Loop

1. **Curie's Law (Paramagnetism):**
$$\chi_m = \frac{C}{T} \implies M = C\left(\frac{B_0}{T}\right)$$
2. **Curie-Weiss Law (Ferromagnetism for $T > T_c$):** Above the critical Curie temperature T_c, a ferromagnetic material abruptly transforms into an ordinary paramagnetic substance:
$$\chi_m = \frac{C'}{T - T_c} \quad (T > T_c)$$
3. **Hysteresis Loop ($B - H$ Curve):**
   - **Retentivity (B_r):** The residual magnetic flux density remaining in a ferromagnetic core when external magnetizing field H is reduced to zero.
   - **Coercivity (H_c):** The reverse magnetizing field required to completely demagnetize the core ($B = 0$).
   - **Electromagnet Cores (e.g., Soft Iron):** Require **high retentivity, low coercivity, and narrow hysteresis loop** to minimize AC hysteresis energy dissipation ($W_h = \int H dB$).
   - **Permanent Magnets (e.g., Steel, Alnico):** Require **high retentivity and high coercivity** to resist demagnetization from stray fields and mechanical shocks.

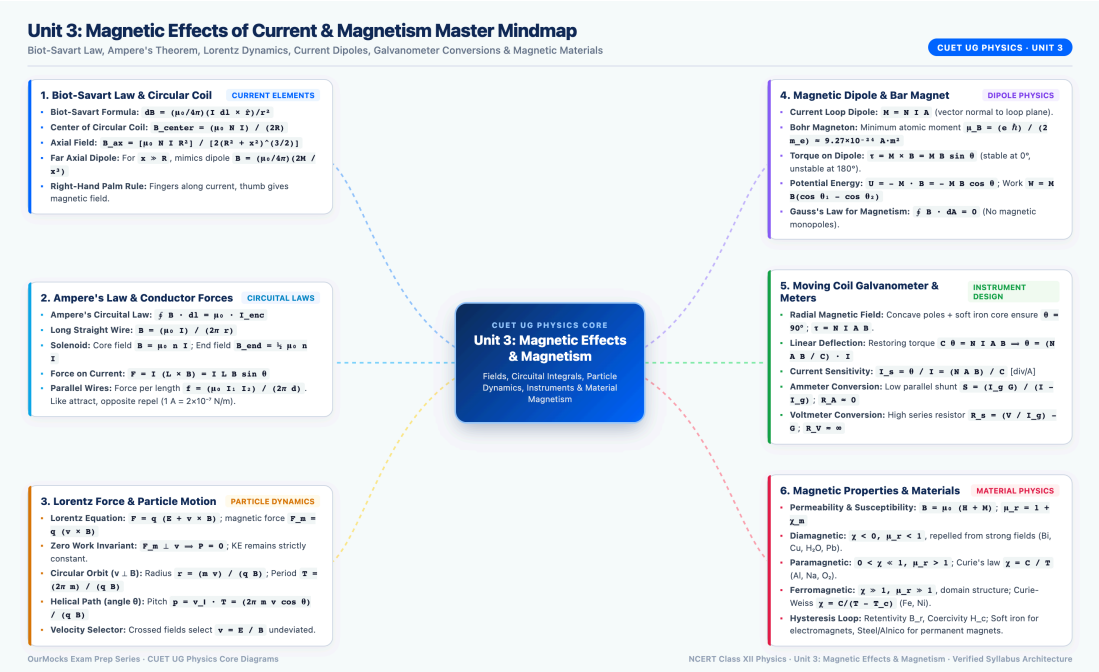


Figure 3: Master Synthesis Mindmap: Complete structural architecture of Unit 3 Magnetic Effects of Current and Magnetism.

MASTER MAGNETIC EFFECTS & MAGNETISM REVISION CHEATSHEET

- **Biot-Savart Law:** $d(B) = \left(\frac{\mu_0}{4}\pi\right)\frac{Id(l)\times\hat{r}}{r^2}$ | Loop center: $B = \frac{\mu_0 NI}{2R}$ | Axial: $B = \frac{\mu_0 NIR^2}{2(R^2+x^2)^{\frac{3}{2}}}$
- **Ampere's Law:** $\oint(B)\cdot d(l) = \mu_0 I_{\text{enc}}$ | Wire: $B = \frac{\mu_0 I}{2\pi r}$ | Solenoid core: $B = \mu_0 nI$, End: $\frac{1}{2}\mu_0 nI$
- **Lorentz Force:** $(F) = q((E) + (v)\times(B))$ | No work: $\Delta K = 0$ | Orbit: $r = \frac{mv}{qB}$ | Period: $T = \frac{2\pi m}{qB}$
- **Forces on Currents:** $(F) = I((L)\times(B))$ | Parallel wires: $f = \frac{\mu_0 I_1 I_2}{2\pi d}$ (like attract, unlike repel)
- **Dipole & Torque:** $(M) = NI(A)$ | $(\tau) = (M)\times(B)$ | $U = -(M)\cdot(B)$ | Bohr magneton: $\mu_B = \frac{e\frac{h}{2\pi}}{2m_e} \approx 9.27\times10^{-24}$ A $\cdot$ m^2
- **Galvanometer:** $\theta = \left(NA\frac{B}{C}\right)I$ | Shunt: $S = \frac{I_g G}{I - I_g}$ | Multiplier: $R_s = \frac{V}{I_g} - G$
- **Magnetic Materials:** $B = \mu_0(H + M)$ | $\mu_r = 1 + \chi_m$ | Dia: $\chi < 0$ | Para: $\chi = \frac{C}{T}$ | Ferro: $\chi = \frac{C'}{T - T_c}$

CUET UG Exam-Style Practice Questions

Q1. A circular coil of radius R carries an electric current I. The magnetic field at a point on the axis at distance $x = \sqrt{3}R$ from the center is related to center field B_0 by:

(A) $B = \frac{B_0}{4}$ **(B)** $B = \frac{B_0}{16}$
(C) $B = \frac{B_0}{8}$ **(D)** $B = \frac{B_0}{2\sqrt{2}}$

Sol. Q1. $B(x) = \frac{\mu_0 I R^2}{2(R^2+x^2)^{\frac{3}{2}}}$. Substituting $x = \sqrt{3}R$ yields denominator $2\left(R^2 + 3R^2\right)^{\frac{3}{2}} = 2\left(4R^2\right)^{\frac{3}{2}} = 2(8R^3) = 16R^3$. Thus $B = \frac{\mu_0 I}{16R} = \frac{1}{8}\left(\frac{\mu_0 I}{2R}\right) = \frac{B_0}{8}$.

Answer: (C)

Q2. A charged particle moves with velocity (v) through a region containing crossed electric and magnetic fields (E) and (B). The particle will pass undeflected if:

(A) $(v) = (E) \times (B)$ **(B)** $(v) = \frac{(E)\times(B)}{B^2}$ and $v = \frac{E}{B}$
(C) $(v) = \frac{(B)\times(E)}{E^2}$ **(D)** $v = \frac{B}{E}$

Sol. Q2. Undeviated condition requires net Lorentz force to be zero: $q(E) + q((v) \times (B)) = 0 \Longrightarrow (E) = -((v) \times (B)) = (B) \times (v)$. Taking cross product with (B) gives $(v) = \frac{(E)\times(B)}{B^2}$ with magnitude $v = \frac{E}{B}$.

Answer: (B)

Q3. An electron and a proton having identical kinetic energies enter perpendicularly into a uniform magnetic field. Which statement is correct regarding their circular paths?

(A) The electron describes a path of larger radius **(B)** The proton describes a path of larger radius
(C) Both particles describe paths of identical radius **(D)** The trajectory of the electron is a straight line

Sol. Q3. Gyroradius $r = \frac{\sqrt{2mK}}{qB}$. Since K and charge magnitude $q = e$ are identical for both, radius is proportional to square root of mass: $r \propto \sqrt{m}$. Since $m_p \approx 1836 m_e$, the proton describes a path with much larger radius ($r_p \gg r_e$).

Answer: (B)

Q4. Two long, parallel straight conductors separated by 10 cm carry currents $I_1 = 4$ A and $I_2 = 10$ A in the same direction. The force per unit length between them is:

(A) 8×10^{-5} N/m, attractive **(B)** 8×10^{-5} N/m, repulsive
(C) 4×10^{-5} N/m, attractive **(D)** 4×10^{-5} N/m, repulsive

Sol. Q4. $f = \frac{\mu_0 I_1 I_2}{2\pi d} = \frac{(4\pi\times10^{-7})\times4\times10}{2\pi\times0.10} = \frac{2\times10^{-7}\times40}{0.10} = 8.0 \times 10^{-5}$ N/m. Since currents flow in the same direction, the force is strictly attractive.

Answer: (A)

Q5. A galvanometer of resistance 50Ω gives full-scale deflection for 10 mA. To convert it into an ammeter reading up to 5 A, the required shunt resistance is approximately:

(A) 0.10Ω **(B)** 0.05Ω
(C) 0.20Ω **(D)** 1.0Ω

Sol. Q5. Shunt formula: $S = \frac{I_g G}{I - I_g} = \frac{0.010\times50}{5-0.010} = \frac{0.50}{4.99} \approx 0.1002\Omega \approx 0.10\Omega$.

Answer: (A)

Q6. If the number of turns in a moving coil galvanometer is doubled to increase its sensitivity, which of the following is true?

(A) Current sensitivity doubles, voltage sensitivity doubles
(B) Current sensitivity doubles, voltage sensitivity remains unchanged
(C) Both current and voltage sensitivity remain unchanged
(D) Current sensitivity halves, voltage sensitivity doubles

Sol. Q6. Current sensitivity $I_s = \frac{NAB}{C} \propto N$, so doubling N doubles I_s. However, doubling wire length doubles coil resistance $G' = 2G$. Thus voltage sensitivity $V_s = \frac{I_s}{G} = \frac{2I_s}{2G} = V_s$ remains unchanged.

Answer: (B)

Q7. A magnetic dipole of magnetic moment (M) is rotated from stable equilibrium ($\theta = 0°$) to unstable equilibrium ($\theta = 180°$) in a uniform field (B). Work done is:

(A) MB
(B) 0
(C) $-2MB$
(D) $2MB$

Sol. Q7. Work done against magnetic field: $W = U_f - U_i = -MB\cos(180°) - (-MB\cos(0°)) = MB - (-MB) = 2MB$.

Answer: (D)

Q8. The magnetic susceptibility of a paramagnetic material at $-73°$C (200 K) is 0.0060. Its susceptibility at $+127°$C (400 K) will be:

(A) 0.0030
(B) 0.0120
(C) 0.0015
(D) 0.0060

Sol. Q8. By Curie's Law, $\chi \propto \frac{1}{T}$ (where T is in Kelvin). $T_1 = 273 - 73 = 200$ K and $T_2 = 273 + 127 = 400$ K. $\frac{\chi_2}{\chi_1} = \frac{T_1}{T_2} = \frac{200}{400} = \frac{1}{2} \Longrightarrow \chi_2 = \frac{0.0060}{2} = 0.0030$.

Answer: (A)

Q9. Which of the following substances exhibits a negative magnetic susceptibility that is independent of temperature?

(A) Aluminum
(B) Sodium
(C) Iron
(D) Copper

Sol. Q9. Diamagnetic materials like copper, bismuth, and water possess negative magnetic susceptibility ($\chi < 0$) resulting from Lenz's law action on paired electron orbits, which is completely independent of temperature.

Answer: (D)

Q10. A closely wound solenoid of 2000 turns and area of cross-section 1.6×10^{-4} m^2 carries a current of 4.0 A. Its associated magnetic dipole moment is:

(A) 1.28 A · m^2
(B) 0.64 A · m^2
(C) 2.56 A · m^2
(D) 12.8 A · m^2

Sol. Q10. Magnetic dipole moment $M = NIA = 2000 \times 4.0 \times (1.6 \times 10^{-4}) = 8000 \times 1.6 \times 10^{-4} = 1.28$ A · m^2 $= 1.28$ J/T.

Answer: (A)

Q11. For making permanent magnets, the material selected should possess:

(A) High retentivity and low coercivity **(B)** Low retentivity and low coercivity
(C) High retentivity and high coercivity **(D)** Low retentivity and high coercivity

Sol. Q11. Permanent magnets require high retentivity (strong remnant magnetization) and high coercivity (to resist stray demagnetizing fields). Typical choices are Steel and Alnico.

Answer: (C)

Q12. An electron enters a region of uniform magnetic field with velocity directed at 30° to field lines. The resulting path is:

(A) A circle **(B)** A straight line
(C) A helix **(D)** A parabola

Sol. Q12. $v_{\parallel} = v\cos 30°$ produces linear translation along (B), while $v_{\perp} = v\sin 30°$ produces transverse circular motion. Superposition yields a helix.

Answer: (C)

Q13. According to Gauss's Law for magnetism ($\oint (B) \cdot d(A) = 0$), which statement must be true?

(A) Magnetic fields cannot penetrate closed surfaces **(B)** Isolated magnetic monopoles do not exist
(C) Current loops produce no external magnetic field **(D)** Magnetic poles have unequal strength

Sol. Q13. Net magnetic flux across any closed surface is zero, confirming magnetic field lines form continuous closed loops; isolated magnetic monopoles do not exist.

Answer: (B)

Q14. A galvanometer with coil resistance $G = 100\Omega$ requires a series resistor $R_s = 4900\Omega$ to read $0 - 50$ V. Full-scale current I_g is:

(A) 1 mA **(B)** 5 mA
(C) 0.1 mA **(D)** 10 mA

Sol. Q14. $V = I_g(G + R_s) \Longrightarrow 50 = I_g(100 + 4900) = 5000 I_g \Longrightarrow I_g = \frac{50}{5000} = 0.010$ A $=$ 10 mA.

Answer: (D)

Q15. When a ferromagnetic material is heated above its Curie temperature T_c, it behaves as a:

(A) Diamagnetic material with negative susceptibility **(B)** Superconductor with zero electrical resistivity
(C) Paramagnetic material with $\chi \propto \frac{1}{T - T_c}$ **(D)** Non-magnetic insulator

Sol. Q15. Above T_c, thermal agitation destroys domain alignment, transforming the ferromagnetic material into an ordinary paramagnetic substance obeying the Curie-Weiss law: $\chi = \frac{C'}{T - T_c}$.

Answer: (C)